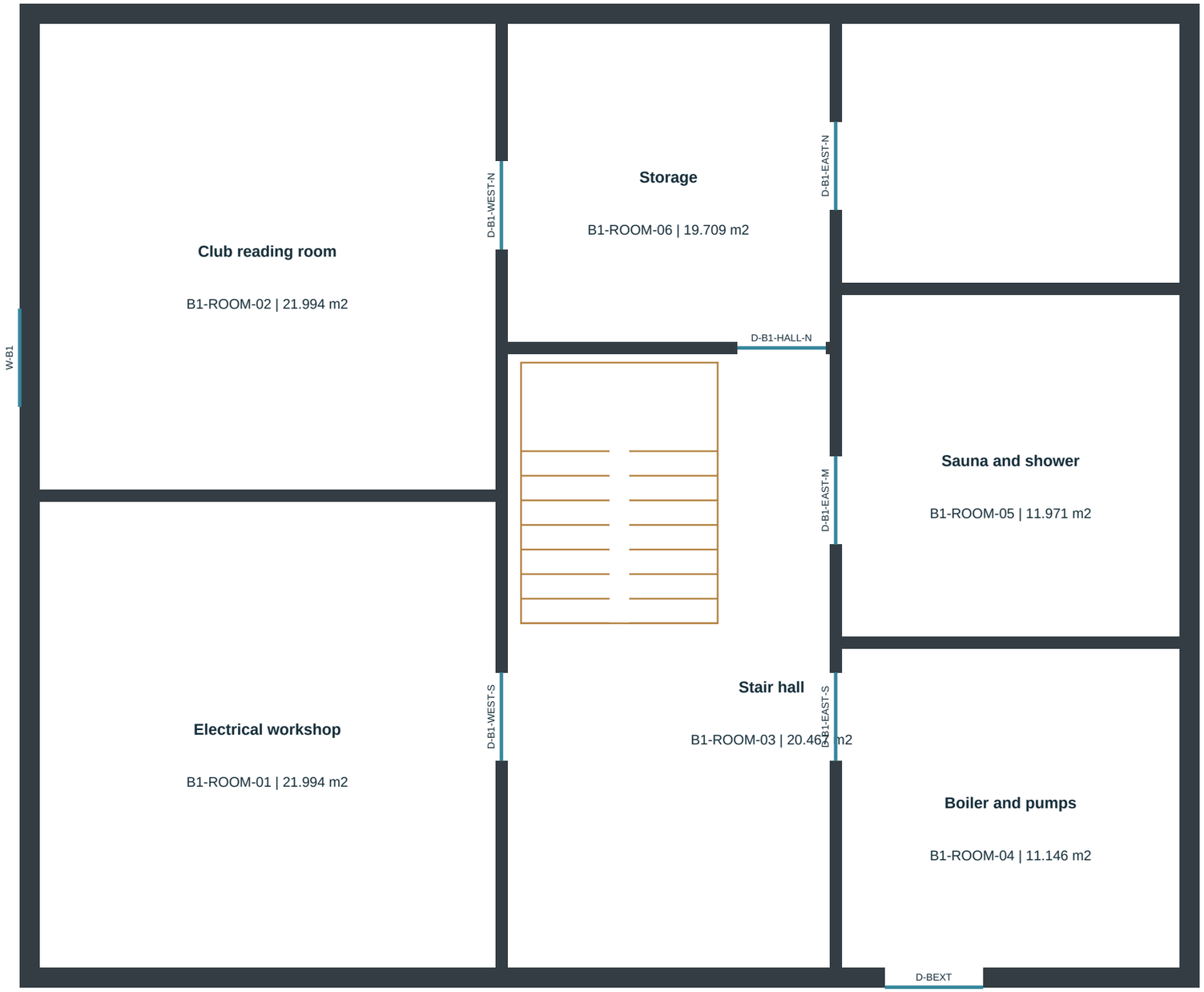
JOHANSSON HOUSE | B1 | COORDINATED PLAN

10 000 mm
overall



12 000 mm overall

FFL -2800 mm

Walls: 200 / 120 mm

Internal apertures: 900 mm

Sliding doors: ID schedule

Stair: 16 x 175 mm

Treads 250; flights 900 mm

Side passage: 1100 mm

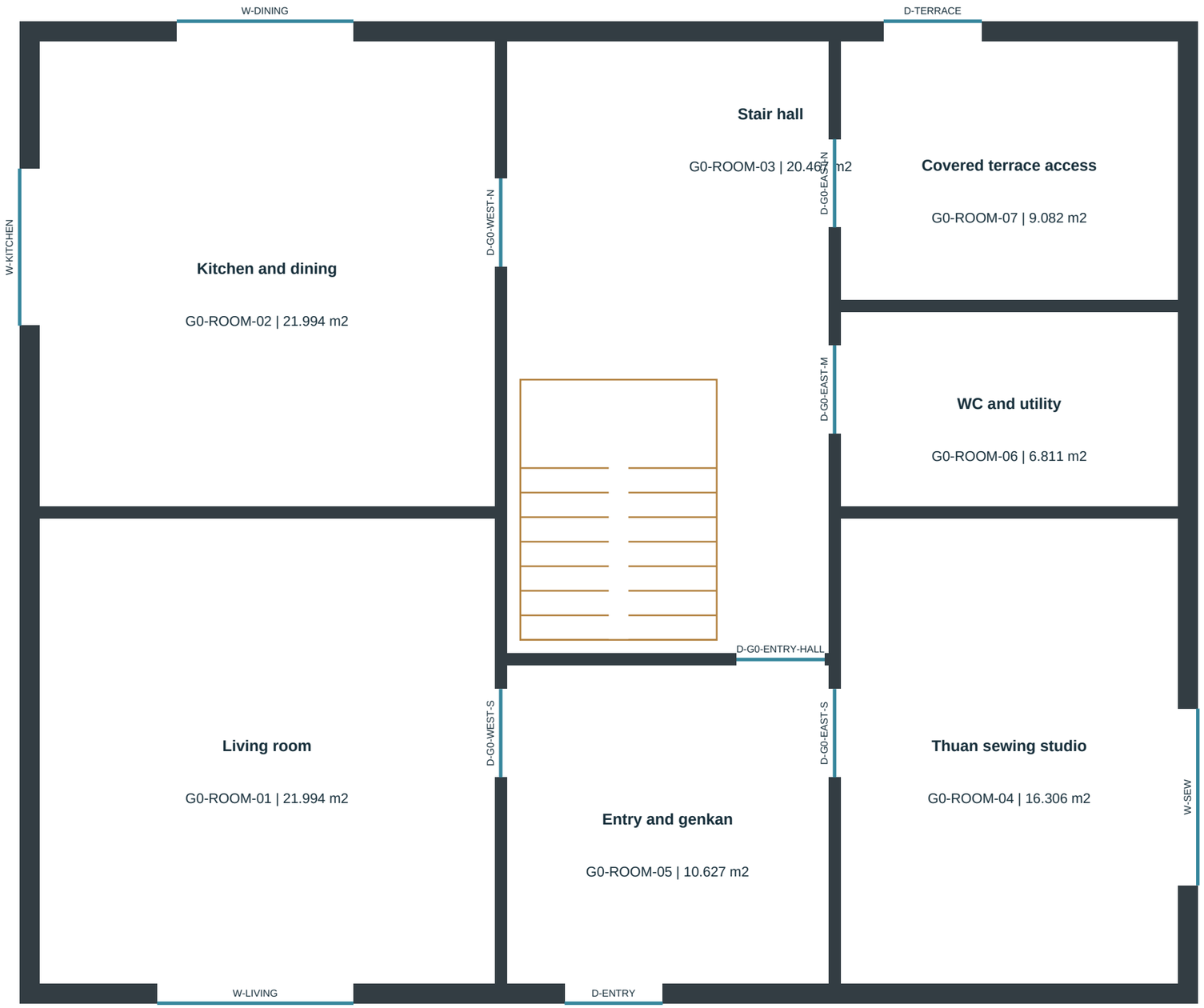
Scale 1:50 at full A3 size

Use CAD for dimensions

North: page up

JOHANSSON HOUSE | G0 | COORDINATED PLAN

10 000 mm
overall



12 000 mm overall

FFL 0 mm

Walls: 200 / 120 mm

Internal apertures: 900 mm

Sliding doors: ID schedule

Stair: 16 x 175 mm

Treads 250; flights 900 mm

Side passage: 1100 mm

Scale 1:50 at full A3 size

Use CAD for dimensions

North: page up

JOHANSSON HOUSE | L1 | COORDINATED PLAN

10 000 mm
overall

12 000 mm overall

FFL 2800 mm

Walls: 200 / 120 mm

Internal apertures: 900 mm

Sliding doors: ID schedule

Stair: 16 x 175 mm

Treads 250; flights 900 mm

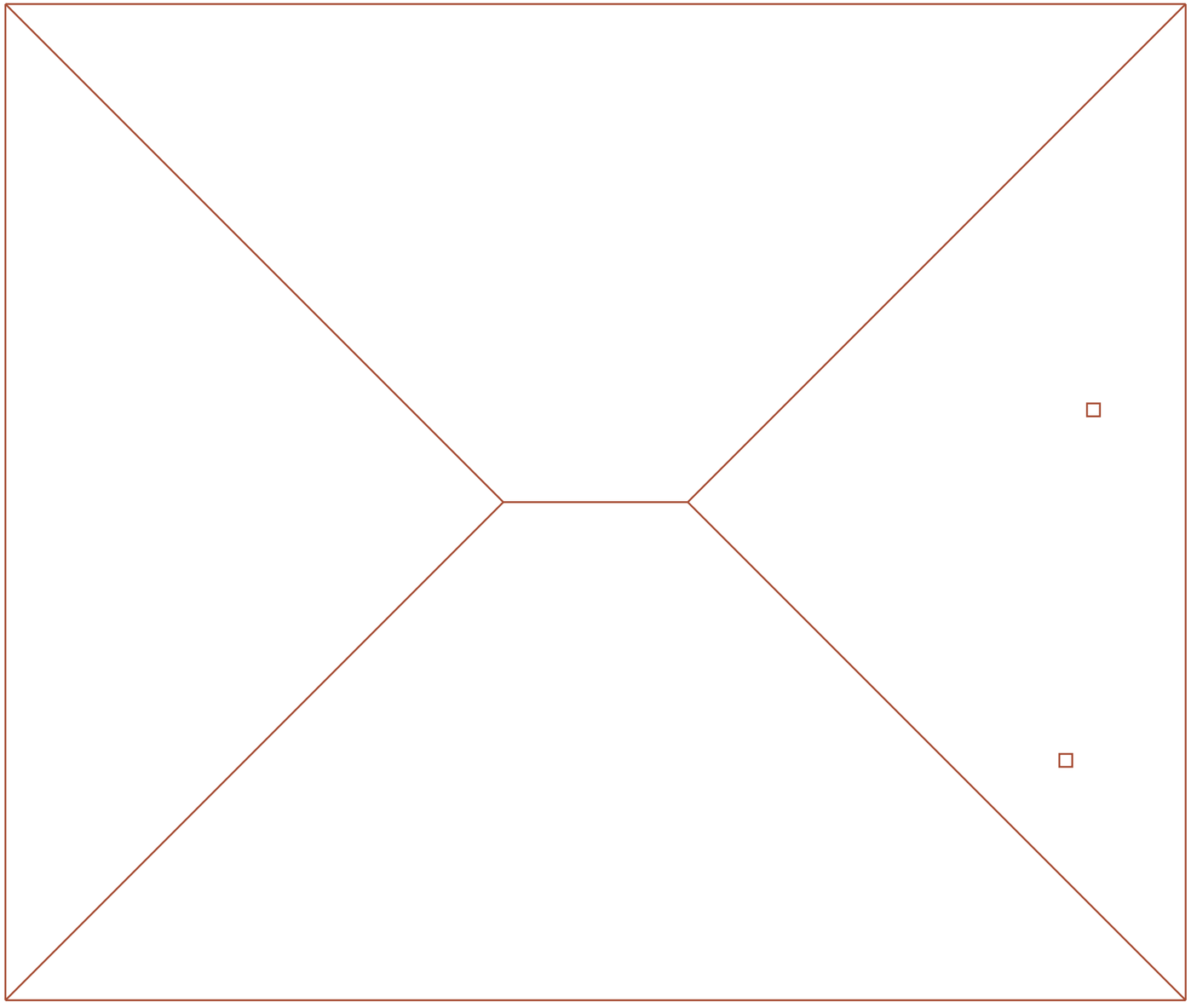
Side passage: 1100 mm

Scale 1:50 at full A3 size

Use CAD for dimensions

North: page up

JOHANSSON HOUSE | A104 | HIPPED ROOF



Eaves: +5800 mm

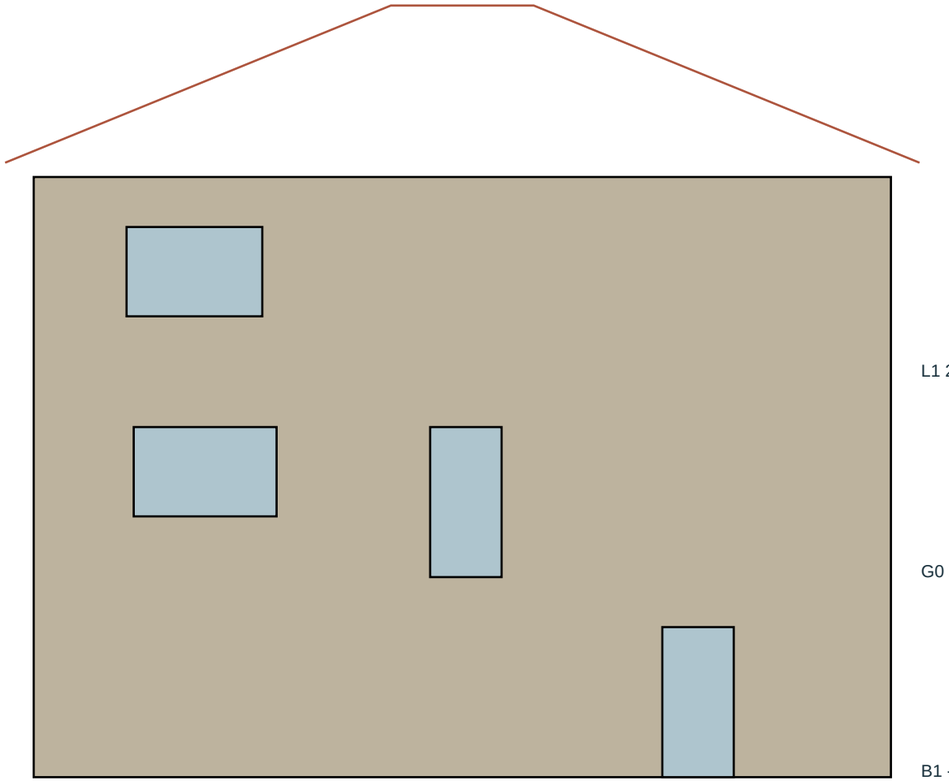
Ridge: +8000 mm

Overhang: 400 mm

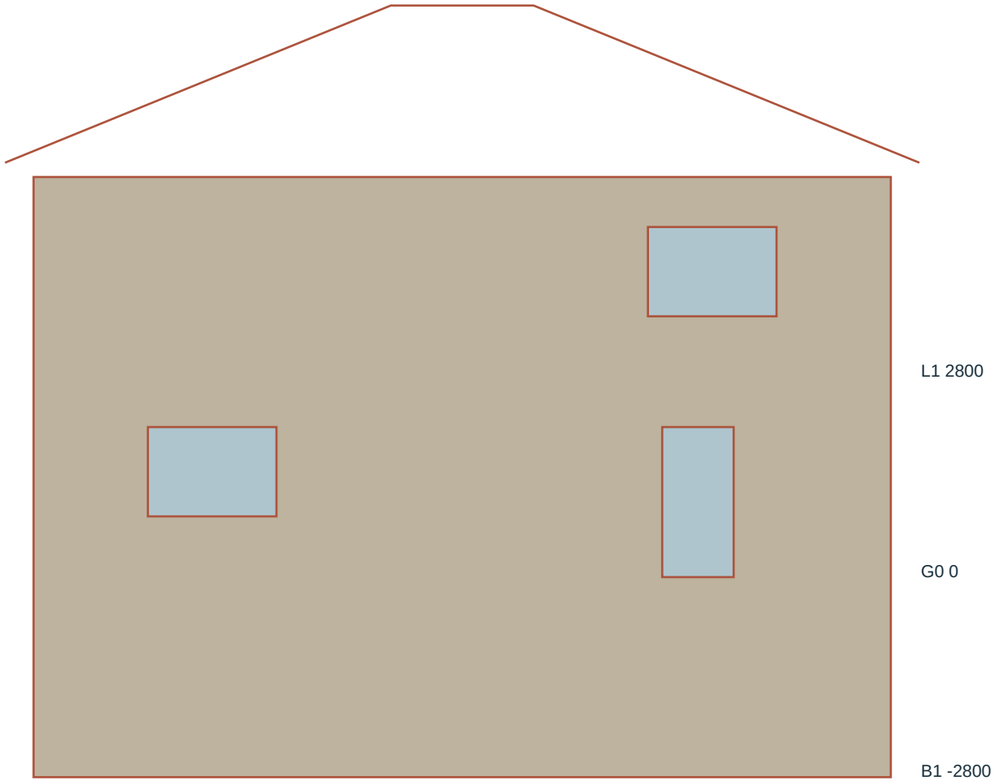
2 downpipes

Roof plan: 138.24 m2

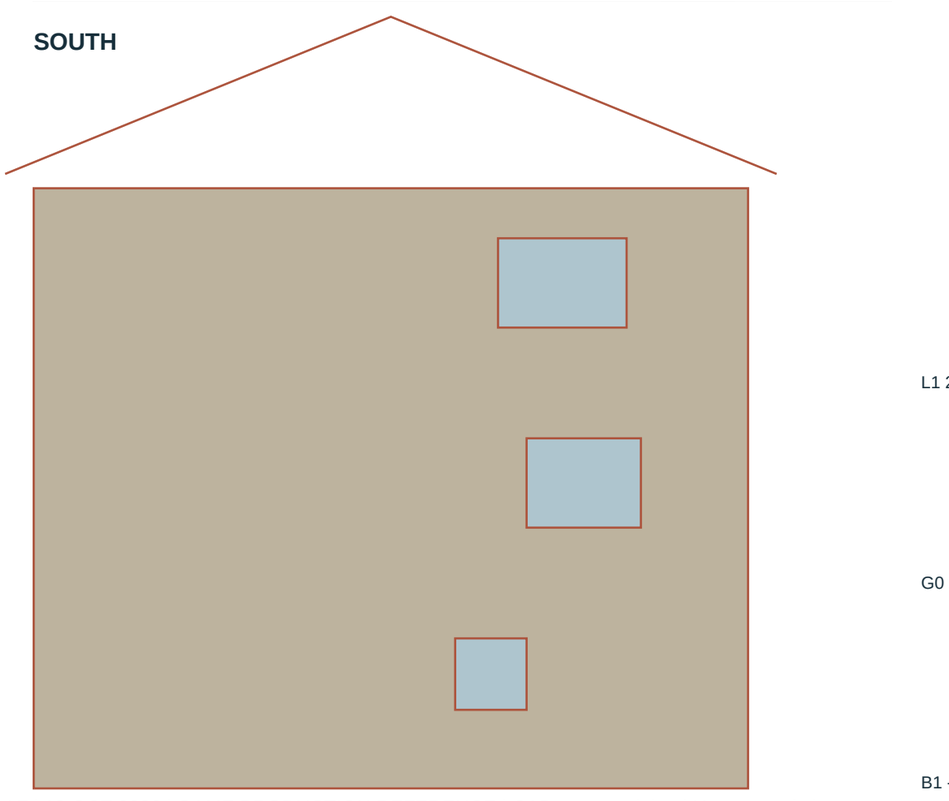
JOHANSSON HOUSE | A201 | ELEVATIONS



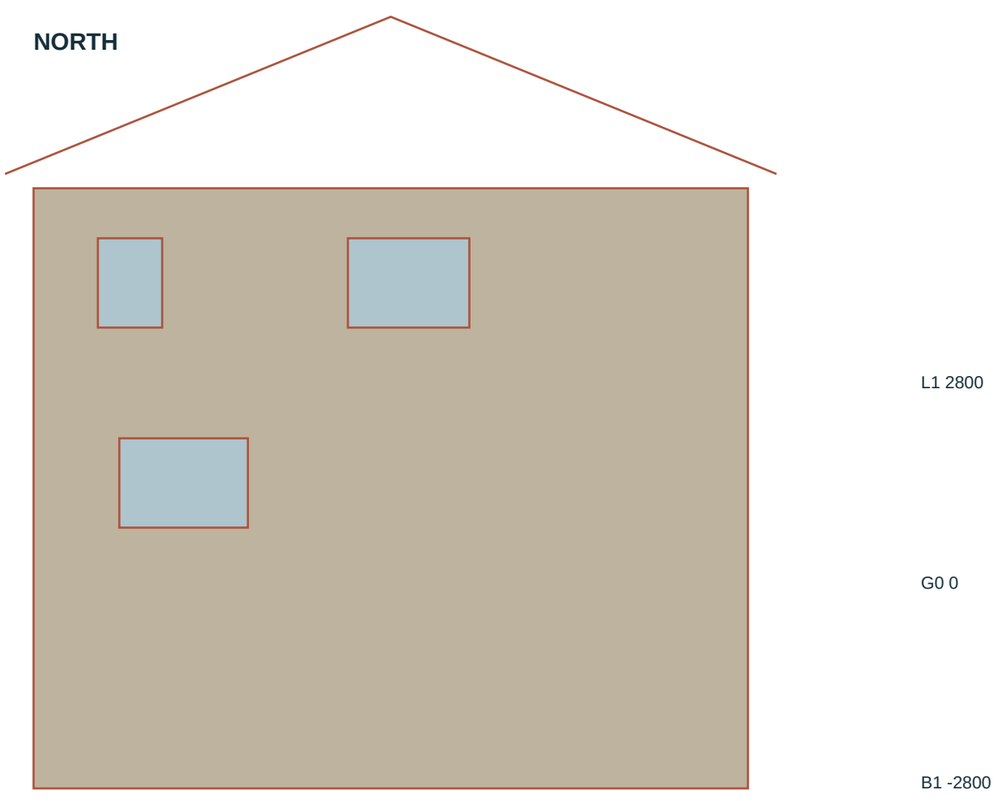
SOUTH



NORTH

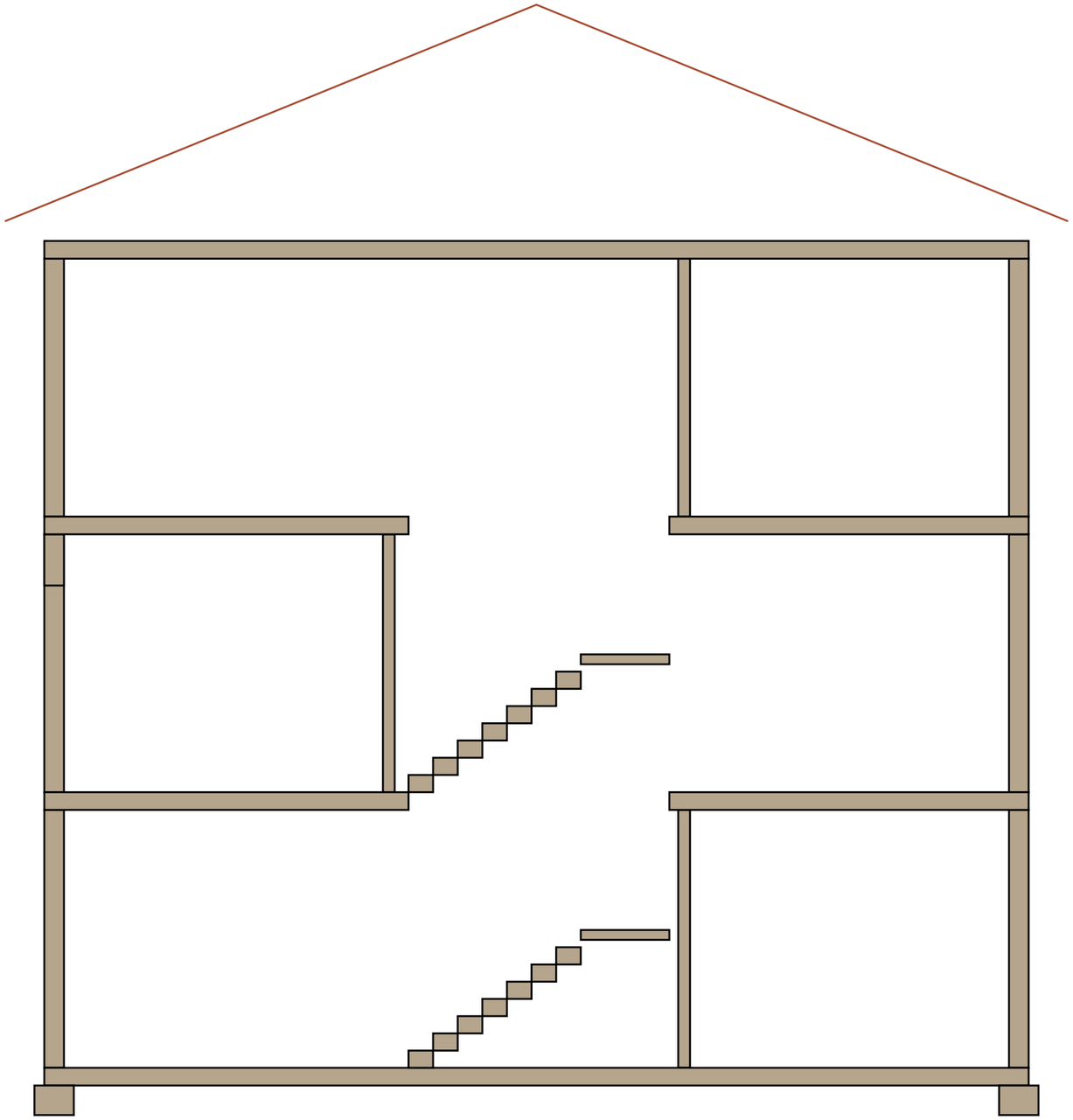


WEST



EAST

JOHANSSON HOUSE | A301 | STAIR SECTION X = 5550



Headroom target: 2000 mm

Slabs: 180 mm

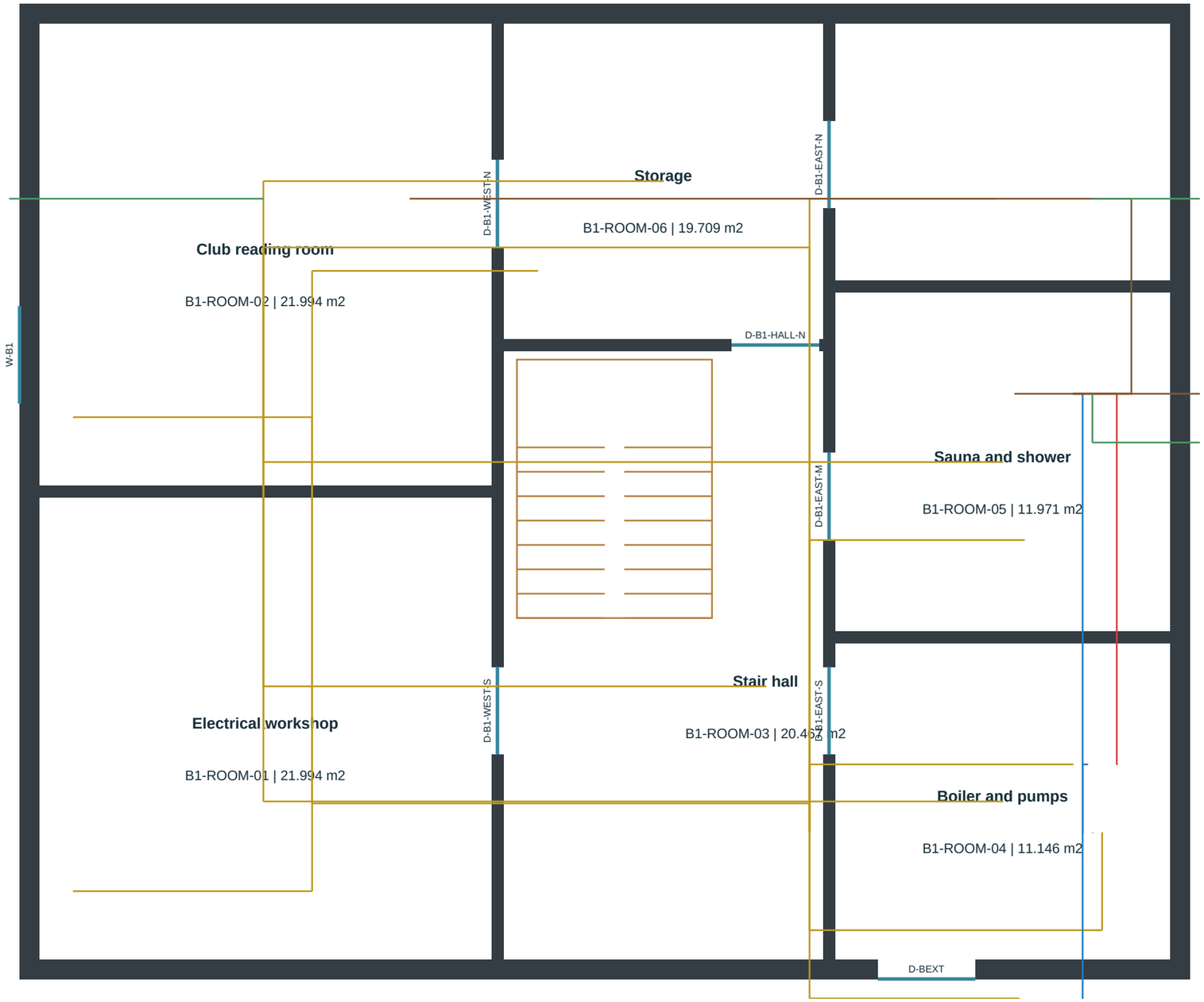
Finished floors: -2800 / 0 / 2800

Roof deck: +5420 to +5600

Stair cut through west flight

JOHANSSON HOUSE | B1 | SERVICE COORDINATION OVERLAY

10 000 mm
overall



12 000 mm overall

FFL -2800 mm

Walls: 200 / 120 mm

Internal apertures: 900 mm

Sliding doors: ID schedule

Stair: 16 x 175 mm

Treads 250; flights 900 mm

Side passage: 1100 mm

Scale 1:50 at full A3 size

Use CAD for dimensions

North: page up

Route colours:

Gold: electrical

Blue/red: cold/hot

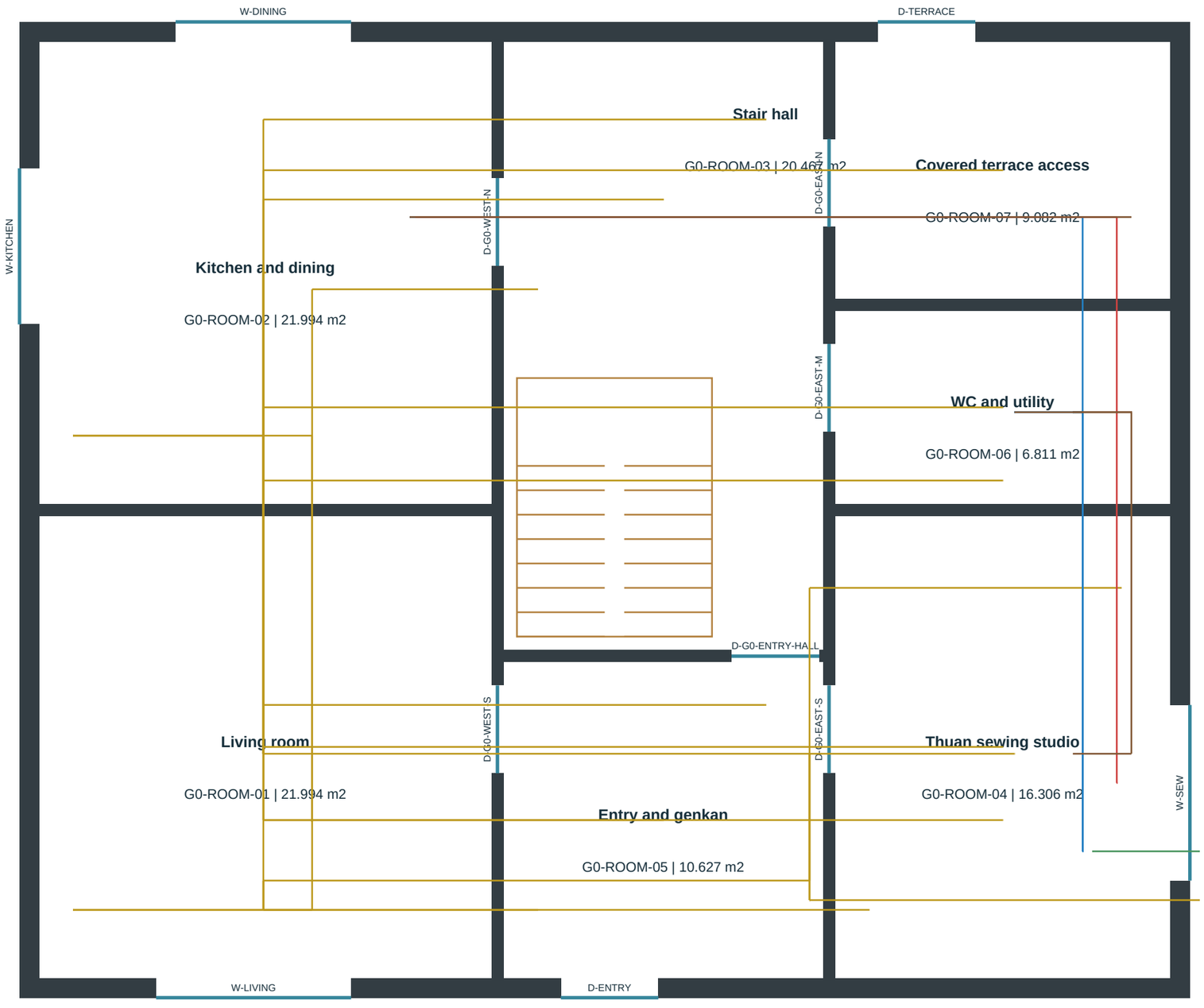
Brown: sanitary

Green: ventilation

Route IDs and Z: CSV/DXF

JOHANSSON HOUSE | G0 | SERVICE COORDINATION OVERLAY

10 000 mm
overall



12 000 mm overall

FFL 0 mm

Walls: 200 / 120 mm

Internal apertures: 900 mm

Sliding doors: ID schedule

Stair: 16 x 175 mm

Treads 250; flights 900 mm

Side passage: 1100 mm

Scale 1:50 at full A3 size

Use CAD for dimensions

North: page up

Route colours:

Gold: electrical

Blue/red: cold/hot

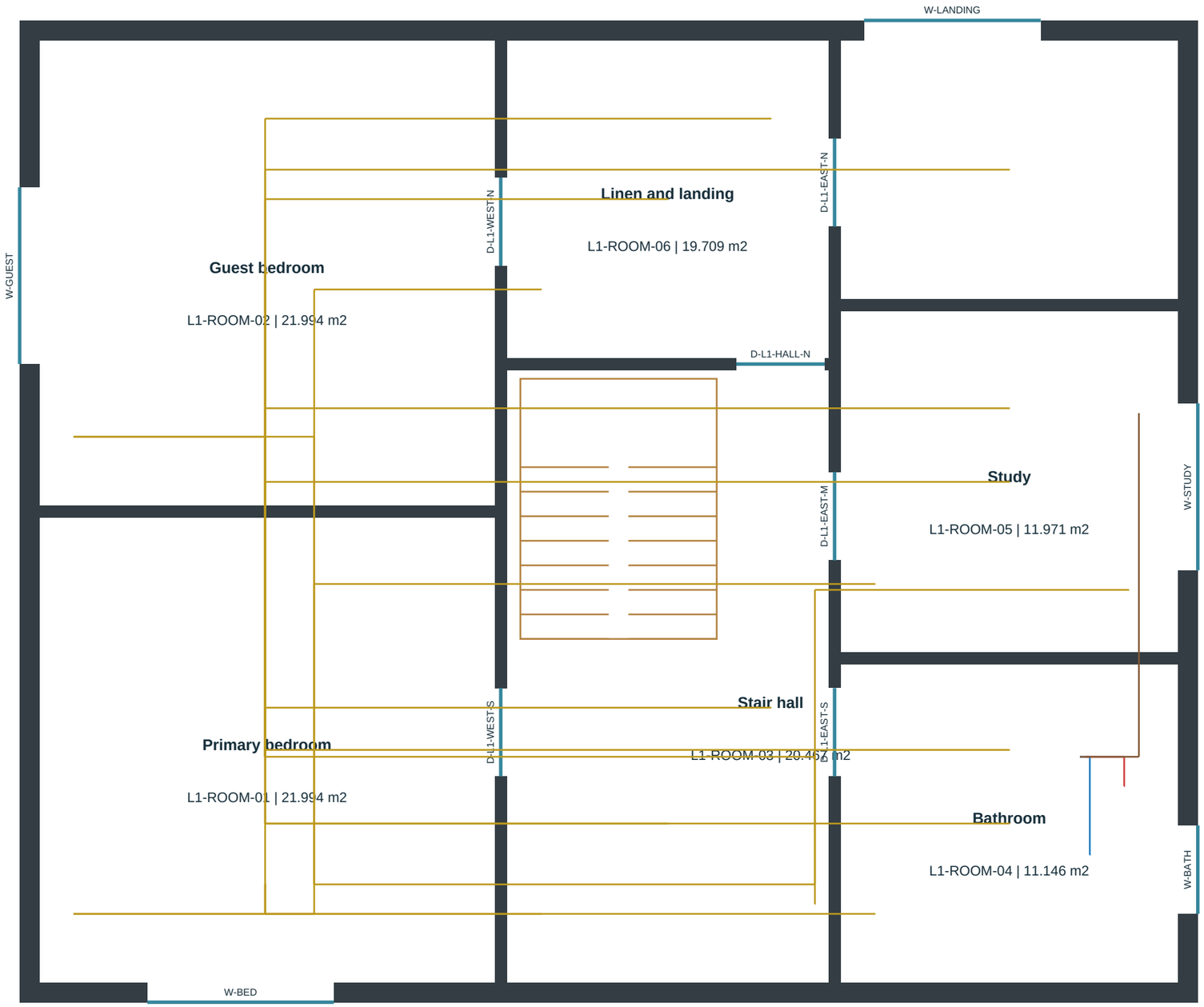
Brown: sanitary

Green: ventilation

Route IDs and Z: CSV/DXF

JOHANSSON HOUSE | L1 | SERVICE COORDINATION OVERLAY

10 000 mm
overall



12 000 mm overall

FFL 2800 mm

Walls: 200 / 120 mm

Internal apertures: 900 mm

Sliding doors: ID schedule

Stair: 16 x 175 mm

Treads 250; flights 900 mm

Side passage: 1100 mm

Scale 1:50 at full A3 size

Use CAD for dimensions

North: page up

Route colours:

Gold: electrical

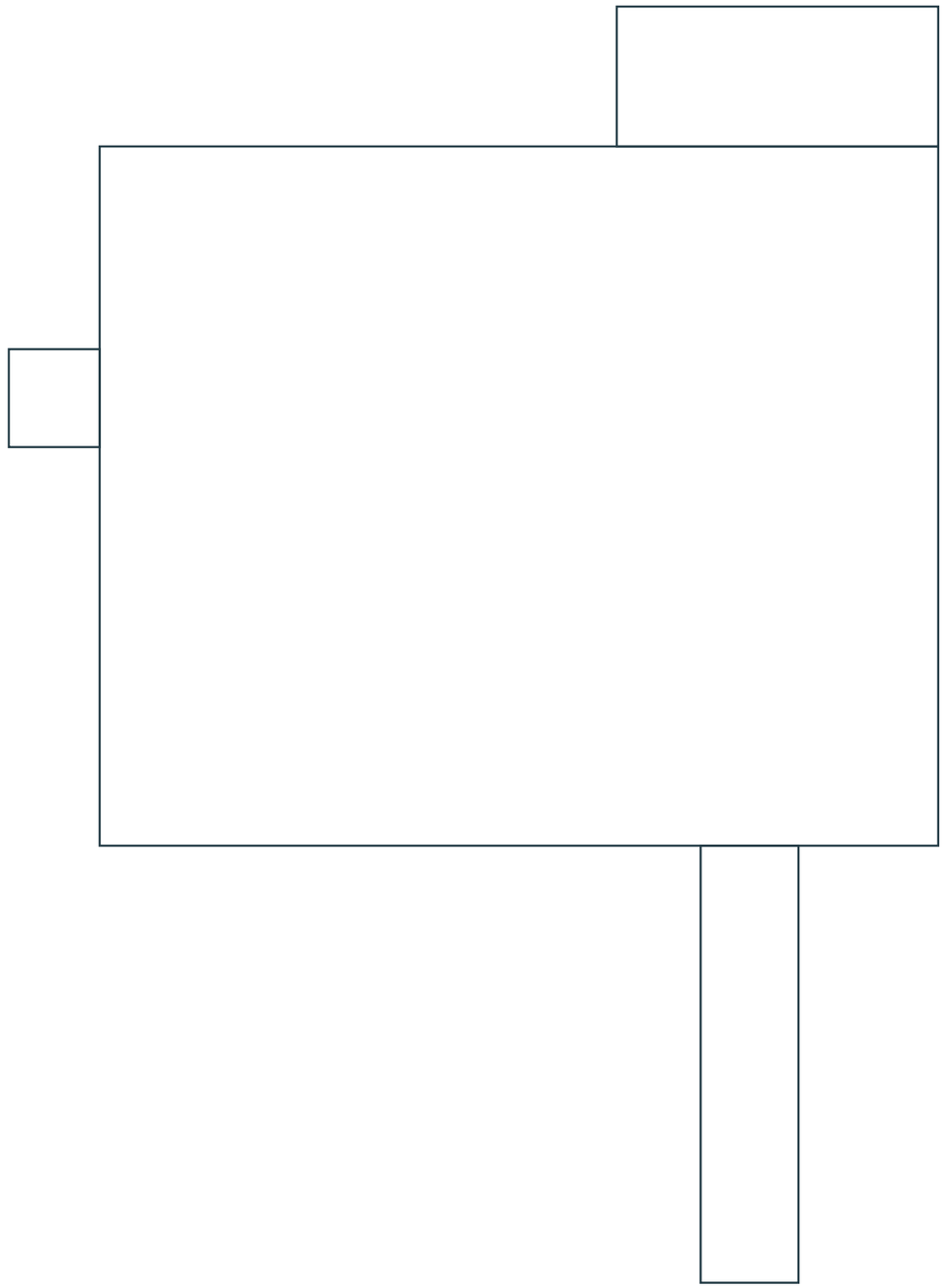
Blue/red: cold/hot

Brown: sanitary

Green: ventilation

Route IDs and Z: CSV/DXF

JOHANSSON HOUSE | C101 | BASEMENT COURT AND SITE



Assumed plot; no survey

Garden stair: 16 x 175 / 250

Clear width: 1000 mm

Court landing: 1400 x 1300

Court and lightwell to storm sump

Sanitary lift pump is independent

Storm/sewer outlets remain separate

Window W-B1 serves lightwell

Waterproofing is a game assembly

POLE -> METER -> MAIN | game 100/200 V, 60 Hz | main demand budget 12 kW

Each branch has an isolation state and game trip threshold; installation cable and protection ratings are outside scope.

B1_DB | BASEMENT

WORKSHOP
1500 W / 100 V / leg L1 / 15.00 A
Game trip budget: 1875.0 W

SAUNA
4500 W / 200 V / leg both / 22.50 A
Game trip budget: 5625.0 W

PUMP
400 W / 100 V / leg L2 / 4.00 A
Game trip budget: 500.0 W

LIGHT_B
180 W / 100 V / leg L2 / 1.80 A
Game trip budget: 225.0 W

VENT
180 W / 100 V / leg L1 / 1.80 A
Game trip budget: 225.0 W

HEATER_E
250 W / 100 V / leg L2 / 2.50 A
Game trip budget: 312.5 W

STORM_PUMP
350 W / 100 V / leg L1 / 3.50 A
Game trip budget: 437.5 W

Room luminaires and sockets inherit branch state.

G0_DB | GROUND

LIGHT_G
180 W / 100 V / leg L1 / 1.80 A
Game trip budget: 225.0 W

SOCKET_G
1500 W / 100 V / leg L2 / 15.00 A
Game trip budget: 1875.0 W

SEWING
300 W / 100 V / leg L1 / 3.00 A
Game trip budget: 375.0 W

AC_G
900 W / 200 V / leg both / 4.50 A
Game trip budget: 1125.0 W

Room luminaires and sockets inherit branch state.

L1_DB | FIRST FLOOR

LIGHT_L
180 W / 100 V / leg L2 / 1.80 A
Game trip budget: 225.0 W

AC_L
900 W / 200 V / leg both / 4.50 A
Game trip budget: 1125.0 W

SOCKET_L
1000 W / 100 V / leg L2 / 10.00 A
Game trip budget: 1250.0 W

Room luminaires and sockets inherit branch state.

Connected and diversified demand calculations are recorded in Validation_Results.json; 19 room luminaires are distributed under the lighting groups.

JOHANSSON HOUSE | P101 | POTABLE WATER

Functional connections below; physical route IDs, ports, sizes and coordinates are in the route schedule and 3D DXFs.

ISLAND_WATER -> METER	underground potable main
METER -> SHUTOFF	service
SHUTOFF -> COLD_MANIFOLD	cold feed
COLD_MANIFOLD -> HEATER	heater supply
COLD_MANIFOLD -> KITCHEN	cold
COLD_MANIFOLD -> G0_WC	cold
COLD_MANIFOLD -> L1_BATH	cold
COLD_MANIFOLD -> LAUNDRY	cold
COLD_MANIFOLD -> SAUNA_SHOWER	cold
HEATER -> KITCHEN	hot
HEATER -> L1_BATH	hot
HEATER -> SAUNA_SHOWER	hot

Cold: 300 kPa source; 0.6 L/s simultaneous game demand. Heater: 180 L, 60 C storage; 40 C mixed.

Branch visual sizes 20 mm; trunk 32 mm. Values define simulation geometry, not an installation specification.

JOHANSSON HOUSE | P102 | SANITARY AND STORM

Functional connections below; physical route IDs, ports, sizes and coordinates are in the route schedule and 3D DXFs.

KITCHEN -> SOIL_STACK	waste
G0_WC -> SOIL_STACK	soil
L1_BATH -> SOIL_STACK	soil
B1_SHOWER -> B1_PUMP	below-grade waste
B1_PUMP -> SOIL_STACK	pumped discharge
SOIL_STACK -> INSPECTION	gravity sewer
INSPECTION -> SEWER_MAIN	island wastewater collection
RAINWATER -> STORM_DRAIN	separate storm system
LAUNDRY -> SOIL_STACK	waste
STORM_SUMP -> STORM_OUT	pumped storm; independent outfall
COURT_DRAIN -> STORM_SUMP	gravity storm
LIGHTWELL_DRAIN -> STORM_SUMP	gravity storm

Sanitary gravity falls >=2%; B1 lift 1 L/s at 4 m game head; storm sump 5 L/s at 5 m.

Roof rain model: 100 mm/h over 138.24 m2 -> 3.84 L/s. Two downpipes provide 6 L/s game capacity.

Storm and sanitary outfalls remain independent; vent and soil stack are one continuous vented pipe.

Functional connections below; physical route IDs, ports, sizes and coordinates are in the route schedule and 3D DXFs.

OUTDOOR_AIR -> B1_FRESH	filtered fresh air
B1_EXHAUST -> B1_OUT	general extract
SAUNA_EXHAUST -> SAUNA_OUT	independent wet extract
SMOKING_EXHAUST -> SMOKE_OUT	independent smoke extract
G0_AC -> OUTDOOR_UNITS	refrigerant
L1_AC -> OUTDOOR_UNITS	refrigerant

Balanced target: 400 m3/h supply = 180 general + 100 sauna + 120 smoke extract.

G0 and L1 cooling each 4 kW for a 50 m2 selected zone at 80 W/m2. Fans and open windows serve other zones.

Extract ducts are independent; condensate drains to an air-gap storm receptor; sealed heater flue exits roof.